

Sequences and Limits

math-foundations / concept 10

1 Setting

Throughout, (X, d) denotes a metric space (concept 09). A *sequence* in X is a function $x : \mathbb{N} \rightarrow X$, written $(x_n)_{n \in \mathbb{N}}$ or simply (x_n) . We study what it means for such a sequence to settle down toward a point of X , both with and without naming the limit in advance.

2 Convergence

Definition 1 (convergence). A sequence (x_n) in X *converges* to $L \in X$ if

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, d(x_n, L) < \epsilon.$$

We write $x_n \rightarrow L$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} x_n = L$.

Remark 2. The integer N is allowed to depend on ϵ . The content of the definition is: *however small a tolerance ϵ you demand, the tail of the sequence eventually sits inside the open ball of radius ϵ around L , and stays there.*

Proposition 3 (uniqueness of limit). *If $x_n \rightarrow L$ and $x_n \rightarrow L'$ in (X, d) , then $L = L'$.*

Proof. Suppose $L \neq L'$ and let $\epsilon = d(L, L')/3 > 0$. Choose N_1 with $d(x_n, L) < \epsilon$ for $n \geq N_1$ and N_2 with $d(x_n, L') < \epsilon$ for $n \geq N_2$. For $n \geq \max(N_1, N_2)$ the triangle inequality gives $d(L, L') \leq d(L, x_n) + d(x_n, L') < 2\epsilon = \frac{2}{3}d(L, L')$, a contradiction. $\square$

3 The Cauchy property and completeness

Definition 4 (Cauchy sequence). A sequence (x_n) in X is *Cauchy* if

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, d(x_m, x_n) < \epsilon.$$

The Cauchy condition speaks only of the sequence's internal behaviour: the terms eventually cluster within any tolerance, without naming a target.

Definition 5 (completeness). A metric space (X, d) is *complete* if every Cauchy sequence in X converges to a point of X .

Example 6. $\mathbb{R}$ with the usual metric is complete; this is one of the defining features of the real line. By contrast, $\mathbb{Q}$ is not complete: the sequence of decimal truncations of $\sqrt{2}$ is Cauchy in $\mathbb{Q}$ but has no rational limit.

4 Convergent sequences are Cauchy

The central theorem of this lesson runs in one direction unconditionally: convergence always implies the Cauchy property. The converse requires completeness.

Theorem 7 (every convergent sequence is Cauchy). *Let (X, d) be a metric space and let (x_n) be a sequence in X . If (x_n) converges, then (x_n) is Cauchy.*

Proof. Suppose $x_n \rightarrow L$ for some $L \in X$. Fix $\epsilon > 0$. We must produce $N \in \mathbb{N}$ such that $d(x_m, x_n) < \epsilon$ whenever $m, n \geq N$.

Apply the definition of convergence with tolerance $\epsilon/2 > 0$: there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies d(x_n, L) < \frac{\epsilon}{2}.$$

Now let $m, n \geq N$. By the triangle inequality (from the metric axioms),

$$d(x_m, x_n) \leq d(x_m, L) + d(L, x_n) < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, the sequence (x_n) satisfies the Cauchy criterion. $\square$

Remark 8. The $\epsilon/2$ split is the canonical move: we have two distances to control ($d(x_m, L)$ and $d(L, x_n)$), each can be made $< \epsilon/2$, and the triangle inequality combines them into a bound of ϵ . The same trick appears in proofs that continuous functions preserve Cauchy sequences, that uniform limits of continuous functions are continuous, and that completions exist.

5 Bolzano–Weierstrass

Theorem 9 (Bolzano–Weierstrass). *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

This is the prototypical compactness statement: boundedness plus closedness in $\mathbb{R}^d$ implies that every sequence has a convergent subsequence, with the limit lying in the set. The notion of *sequential compactness* (concept 11 and beyond) is the abstract generalisation.

6 Why the Cauchy criterion is useful

In applied work — and especially in numerical analysis and machine learning — we often cannot exhibit a limit in closed form, but we can monitor distances between successive iterates. Theorem ?? together with completeness of $\mathbb{R}^d$ tells us that *shrinking pairwise distances suffice*: if the iterates of an optimisation algorithm satisfy $\|x_m - x_n\| \rightarrow 0$ as $m, n \rightarrow \infty$, then they converge to a point of $\mathbb{R}^d$, regardless of whether we can name that point. This is the foundation of every fixed-point convergence proof, including Picard–Lindelöf (concept 30) and the construction of strong solutions of SDEs (concept 33).